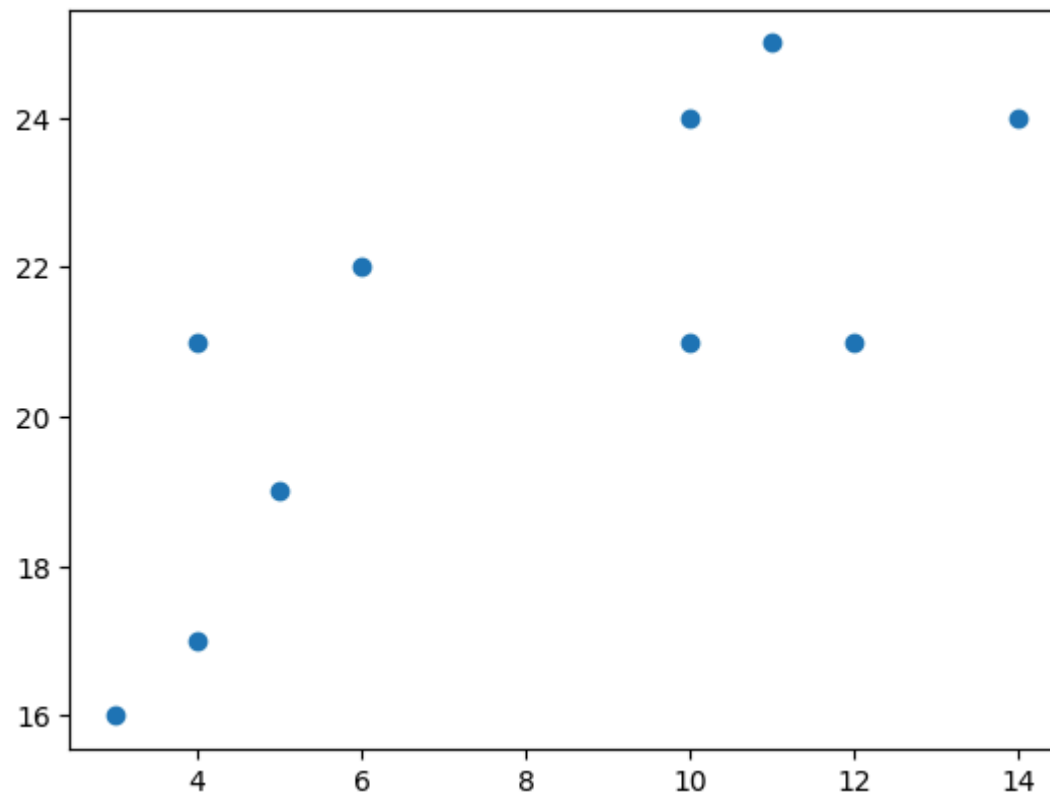


```
In [1]: import numpy as np
import matplotlib.pyplot as plt
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.cluster import AgglomerativeClustering
```

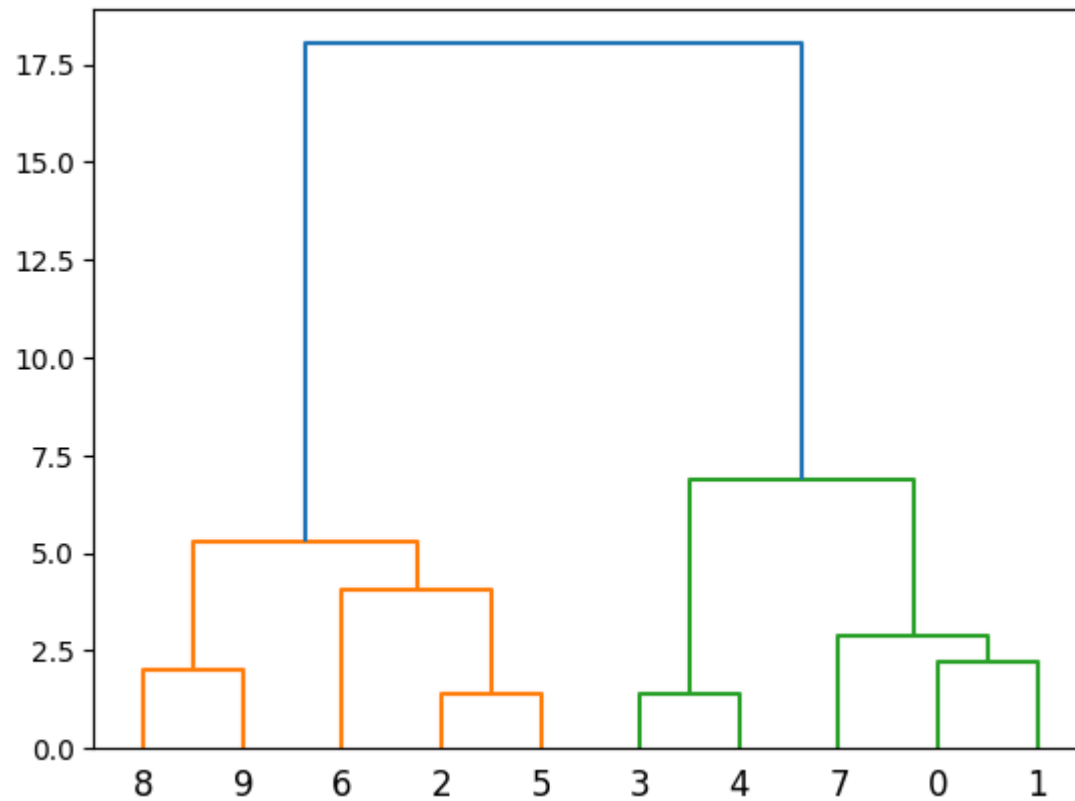
```
In [2]: x = [4,5,10,4,3,11,14,6,10,12]
y = [21,19,24,17,16,25,24,22,21,21]
```

```
In [3]: plt.scatter(x,y)
plt.show()
```



```
In [6]: data = list(zip(x,y))
linkage_data = linkage(data, method='ward' ,metric='euclidean')
```

```
dendrogram(linkage_data)
plt.show()
```



```
In [7]: hierarchial_cluster = AgglomerativeClustering(n_clusters=2,metric='euclidean',linkage='ward')
labels=hierarchial_cluster.fit_predict(data)
print(labels)
```

```
[0 0 1 0 0 1 1 0 1 1]
```

```
In [ ]:
```